

TOWNSVILLE, Australia

WMO: 942940

Lat: 19.2483S

Lon: 146.7661E

Elev: 9

StdP: 101.22

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	11.9	-4.1	2.7	21.5	-0.3	3.7	20.4	10.2	23.2	9.4	23.4	0.9	170	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.4	33.2	24.4	32.2	24.6	31.6	24.5	27.0	30.3	26.5	29.8	26.1	29.4	5.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.5	29.0	25.7	20.9	28.7	25.1	20.3	28.2	85.3	30.8	83.3	30.0	81.4	29.6	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	9.2	8.5	DB	7.5	36.4	1.4	2.5	6.5	38.2	5.7	39.7	4.9	41.1	3.9	42.9	(4)
(5)			WB	5.1	28.1	1.2	0.6	4.2	28.6	3.5	28.9	2.9	29.2	2.0	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.5	28.0	27.9	27.2	25.4	22.9	20.6	19.7	20.4	22.8	25.2	26.7	28.0
	DBStd	3.49	1.40	1.46	1.20	1.40	1.94	2.25	2.21	2.02	1.81	1.56	1.31	1.41
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	25	0	0	0	0	1	7	12	6	0	0	0	0
	CDD10.0	5308	559	502	532	462	398	318	301	322	384	470	502	558
	CDD18.3	2291	301	269	274	212	141	74	54	69	134	212	252	299
	CDH23.3	21023	3432	2995	2790	1736	861	341	234	357	848	1650	2421	3359
CDH26.7	6321	1282	1104	911	456	125	20	6	18	119	366	690	1225	
Wind	WSAvg	4.3	4.7	4.4	4.3	4.2	3.7	3.6	3.6	3.9	4.3	4.9	5.1	5.0
Precipitation	PrecAvg	1069	244	273	157	50	30	19	12	12	10	25	62	140
	PrecMax	2029	967	989	683	406	181	110	102	54	81	253	359	458
	PrecMin	388	15	4	0	1	0	0	0	0	0	0	0	9
	PrecStd	433	211	237	160	69	39	26	21	16	18	40	73	119
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	35.2	33.0	31.4	29.9	28.2	27.3	28.2	30.2	32.2	33.1	34.0
		MCWB	25.3	23.6	24.5	22.5	21.9	20.3	18.7	17.2	18.0	19.7	21.6	24.6
	2%	DB	32.9	33.0	31.9	30.7	28.9	27.1	26.2	27.0	28.9	30.2	31.2	32.6
		MCWB	25.5	25.3	24.6	22.9	21.0	19.9	18.7	18.9	20.1	22.4	23.5	24.5
	5%	DB	31.9	32.0	31.1	29.9	28.0	26.2	25.6	26.1	28.0	29.2	30.4	31.8
		MCWB	25.1	25.2	24.4	22.6	20.7	19.3	18.4	18.6	20.3	22.1	23.2	24.5
	10%	DB	31.1	31.1	30.4	29.1	27.2	25.3	24.8	25.3	27.1	28.7	29.9	31.0
		MCWB	24.7	25.1	24.1	22.3	20.3	18.7	17.9	18.2	20.0	21.8	23.0	24.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	27.6	27.2	25.9	24.1	23.0	22.2	22.5	23.8	25.2	26.2	27.2
		MCDB	31.2	30.7	30.2	28.8	27.4	25.2	24.6	25.1	26.7	28.4	29.1	30.8
	2%	WB	26.8	27.0	26.4	25.1	23.1	21.8	21.1	21.5	22.8	24.4	25.3	26.5
		MCDB	30.4	30.1	29.5	28.1	26.3	25.0	24.0	24.4	26.2	27.7	28.6	30.0
	5%	WB	26.4	26.6	25.9	24.3	22.4	21.0	20.4	20.7	22.2	23.7	24.7	26.0
		MCDB	29.8	29.7	29.1	27.5	25.9	24.4	23.4	24.0	25.7	27.2	28.3	29.4
	10%	WB	26.0	26.1	25.4	23.7	21.8	20.3	19.6	19.8	21.5	23.1	24.2	25.5
		MCDB	29.3	29.3	28.6	27.1	25.6	23.8	22.9	23.5	25.3	26.8	28.0	29.1
Mean Daily Temperature Range	5% DB	MDBR	6.4	6.3	6.9	8.1	9.3	9.8	10.7	10.6	9.4	7.7	6.7	6.4
		MCDBR	7.2	7.4	7.6	8.7	9.6	10.0	10.7	10.9	9.8	7.9	7.1	6.9
		MCWBR	2.8	2.9	2.9	3.4	4.0	4.6	5.0	4.9	4.2	3.1	2.7	2.6
	5% WB	MCDBR	6.4	6.4	6.2	6.9	7.9	8.3	8.1	8.4	7.6	7.0	6.5	6.2
		MCWBR	2.8	3.0	2.8	3.1	3.7	4.0	4.1	4.2	3.4	3.0	2.9	2.7
Clear-Sky Solar Irradiance	taub	0.406	0.398	0.381	0.365	0.350	0.335	0.323	0.338	0.368	0.389	0.392	0.405	
	taud	2.512	2.541	2.578	2.606	2.601	2.625	2.645	2.580	2.491	2.461	2.493	2.481	
	Ebn at Noon	939	936	929	907	884	880	904	920	926	934	946	940	
	Edh at Noon	114	110	102	93	86	81	82	93	109	117	116	118	
All-Sky Solar Radiation	RadAvg	6.32	6.00	5.58	5.03	4.31	3.87	4.33	5.20	6.21	6.83	6.95	6.87	
	RadStd	0.57	0.69	0.53	0.47	0.34	0.35	0.33	0.36	0.39	0.43	0.73	0.68	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	+1.71	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page